

# Hrishika Roychoudhury

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## EDUCATION

**University of North Carolina at Chapel Hill (UNC-CH)** | School of Data & Information Sciences (SDIS) | Chapel Hill, NC

M.S. in **Information Science (MSIS)** | BS-MS Dual Degree Pathway Program

Jan 2026 - Exp. May 2027

B.S. in **Information Science and Neuroscience**, Minor in **Computer Science**

Aug 2022 - Dec 2025

**Relevant Coursework:** Human Information Interactions; AI Literacy; Database Concepts & Applications; AI-Driven Smart Connected Health; Information Systems Analysis & Design; Distributed Knowledge Graphs; Human Factors in Health & AI; Neuropsychopharmacology; Organic Chemistry II; Machine Learning for Computational Biology; Health App Development.

**North Carolina School of Science and Mathematics (NCSSM)** | Durham, NC

Aug 2020 - May 2022

## EXPERIENCE

**Data Science Graduate Teaching Assistant** – UNC School of Data and Information Sciences.

Dec 2025 - Present

- Help teach DATA 120: Ethics of Data Science & AI, guiding **90+** students across twice-weekly class sessions; supervise course activities; hold office hours to address student questions and reinforce complex concepts.
- Design personalized learning experiences to meet every student's needs; grade classwork and maintain records in learning management system; provide constructive individualized feedback on assignments to support academic growth.

**Apprentice** – Girls Who Code × Pluralsight Work Prep Apprenticeship

May 2023 - Jun 2023

- Conducted Pluralsight workshops in full-stack web & software development, Python, Java, and cloud technologies. Practiced with **SDLC**, **Agile** & **Waterfall** methodologies, **Git**, **DevOps**, and **CI/CD** pipelines.
- Handpicked as Closing Ceremony speaker out of **100+** participants; moderated official Discord server for students.

## TECHNICAL SKILLS

**Languages:** English (Fluent), Bengali (Fluent), Spanish (Proficient), Hindi (Novice).

**Programming & frameworks:** SQL, Python, HTML/CSS, Java, JavaScript, React, FHIR, Git, MATLAB, Node.js, Flutter, AJAX.

**Software & collections:** LibGuides, Zotero, Covidence, GitHub Pages, WordPress, Wix, Figma, Adobe (Photoshop, Acrobat DC).

**Productivity:** Microsoft 365 (Excel, Word, PowerPoint, Teams), Google Suite, Zoom, Notion, Obsidian.

## RESEARCH & PROJECTS

Further write-ups and case studies: [hrris28.github.io/garden.html](https://hrris28.github.io/garden.html)

**Eye-Tracking Engagement in AI-Assisted Academic Search** – Supervisor: Dr. Fei Yu.

Sept 2025 - Present

- Designed a controlled between-subjects human-subjects study evaluating how students seek, interpret, and verify academic information using AI tools vs. unaided UNC Library Search (**EBSCOhost/Articles+**).
- Paired **Tobii Nano Pro** eye-tracking with think-aloud protocols, using gaze signals (fixation, saccade, pupil dilation) as cognitive-validity measures of visual attention. Applied behavioral metrics and post-task survey instruments to model cognitive load during information-seeking tasks and inform AI guidelines in education.

**AI Scraping Access Policy Framework** – Supervisors: Will Cross & Dave Hansen.

Jan 2026 - May 2026

- Authored a board-ready annotated policy memo for the **Internet Archive** governing **bulk automated access to digital collections**, weighing interests of publishers, researchers, and AI developers across a stakeholder scenario analysis.
- Wrote Terms of Service text, tiered access framework, model Data Trust Agreement, and staff implementation guidance.

**Web Archive Forensics Service Proposal** – Supervisor: Leslie Thomson.

Mar 2026 - May 2026

- Created a three-tier reference and capture service for the State Library of NC, with **automated web capture**, **metadata extraction**, and **verification workflows** grounded in a statewide information-needs analysis of independent journalists; scoped feasibility, limitations, and library-newsroom service precedents.
- Assessed preservation strategies and institutional capacity against **Archive-It** infrastructure and **WARC/ISO 28500** standards to support technical infrastructure for long-term access to sustainable digital collections.

**LLM-Driven Language Shift in Naturalistic Communication** – Independent Study.

Aug 2025 - Feb 2026

- Executed a **PRISMA**-guided systematic review (6 bibliographic databases, 4 preprint servers) with dual citation chasing, synthesizing evidence on how large language models have affected human communication.
- Drafted a **five-lens conceptual extraction framework** classifying how studies operationalize LLM exposure and measure linguistic outcomes, surfacing gaps and methodological weaknesses alongside a preliminary evidence map.

**Brain-Computer Interface (BCI) Model Card** – Supervisor: Dr. Raghavendra P Pothukuchi.

Jul 2025 - Dec 2025

- Tested end-to-end ML pipeline in TensorFlow/Keras/MNE on neural signal data. Benchmarked three **CNN** architectures (**EEGNet**, **DeepConvNet**, & **ShallowConvNet**), generating performance metrics and training curves.
- Evaluated DREAMS framework HTML model cards with a 13-item **System Explainability Scale (SES)** across Trust, Usability, and Understandability/Interpretability dimensions. Published reproducible artifact to GitHub.

**Health Chart App** – Supervisor: Jonathan Tweedy.

*Sept 2025 - Dec 2025*

- Built a JavaScript Electronic Health Record (EHR) dashboard (**Leaflet.js, AJAX**) using the **FHIR (Fast Healthcare Interoperability Resources) API** for data integration with hospital systems like EPIC, visualizing patient vitals/trends.

**3D-Printed Receptor Model** – Supervisor: Dr. Alexandra Nowlan.

*Aug 2025 - Dec 2025*

- Designed and 3D-printed an interactive D3 dopamine receptor model from cryo-EM structures (**ChimeraX, Blender, CAD**), demonstrating conformational change triggered by a bitopic ligand (FOB02-04A).
- Presented at UNC BeAM Makerfest as a communication tool for medicinal chemists and clinicians to visualize D3 -selective bitopic agonist activity relevant to Parkinson's disease therapeutics.

**Biomedical & Environmental Research** (selected)

*2019 - 2022*

- Synthesized biogenic silver nanoparticles with agri-food byproducts as a novel anticancer reducing agent (Regeneron STS Top 300; NC International Science Challenge); developed chitosan-insulin nanoparticles for oral insulin delivery in noninvasive diabetes treatment (International BioGENEius Challenge); studied algae-based wastewater bioremediation (**published**, International Journal of High School Research; Intel ISEF).
- Employed computational chemistry molecular modeling techniques with **Schrödinger Maestro, Gaussian, MATLAB, Vortex**, and **WebMO**, alongside wet lab research for physicochemical property analyses and medical drug design.

## ACTIVITIES & LEADERSHIP

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**Facilitator & College Loop President** – Girls Who Code (GWC), UNC Chapter

*Chapel Hill, NC | Jan 2023 - Present*

- Developed lesson plans and taught **Python** programming to **50+** students weekly, beginning with **Turtle** for visual, low-barrier learning; lectured, held office hours, led instruction through semester-long app design projects.
- Started UNC's **GWC College Loop (2026)**; guided coding projects and partnered with local schools to expand CS access for underrepresented groups; communicated regularly with parents, alumni, and sponsors.

**Co-President** – UNC Zero Degrees

*Chapel Hill, NC | Jul 2025 - Present*

- Grew UNC's largest interdisciplinary student organization to **200+** members; secured **\$1,800** in Undergraduate Senate Funding for programming and ran community events. Work with board to foster a welcoming environment.
- Managed communication platform, server infrastructure, and hosted bi-weekly activities; configured automated moderation tools, interactive bots, AlertCarolina webhooks, and event analytics, increasing student engagement by **85%**.

**Ambassador & Student Advisory Board Member** – UNC Information & Library Science (ILS)

*Chapel Hill, NC | Aug 2025 - Jun 2026*

- Selected to represent SDIS for prospective students and families through outreach events and academic panels.
- Served on the **ILS New School Student Advisory Board**. Advised institutional decision-making, providing input to school leadership on curriculum, student experience, spaces, and resources across graduate and undergraduate programs.

**Co-Founder & Chief Operating Officer** – SkyeLabs Innovation Inc.

*Durham, NC | Sept 2020 - Aug 2024*

- Directed product development and organizational operations for sustainable materials startup developing **EcoGel**, a flexible biopolymer aerogel with applications in high-performance apparel, aerospace infrastructure, and space exploration. Led team through R&D, funding strategy, and prototyping phases.
- Secured **\$2000** in research grant funding from the Bowman-Brockman Endowment for Entrepreneurship and Advanced Research. Won **Pete Conrad Scholar** in Aerospace & Aviation (International Conrad Spirit of Innovation Challenge 2021, highest honor); recognized as **Top 10 Finalist** at **NASA iTech** & sole Honorable Mention as the youngest team nationally.

**Editor-in-Chief** – Broad Street Scientific (BSS)

*Durham, NC | Oct 2020 - Jun 2022*

- Led annual **100-page** peer-reviewed publication; mentored **20+** editors and writers. Decided theme (Chaos Theory).
- Designed submission management system and website with **SEO Optimization**. Organized publishing workshops; reviewed and critiqued submissions. Published NCSSM-Durham's official journal of student STEM research successfully.

**Community Service**

*Charlotte, NC | 2018 - 2022*

- Taught **16** students in Intro to Chemistry course through SMATH Tutors, supporting **\$24,000** raised for the Food Bank of Central & Eastern NC. Volunteered in **Hematology/Oncology** unit at Novant Health Presbyterian Medical Center. Assisted international students with class integration as an **ESL Ambassador**, translating official documents into Bengali and tutoring English one-on-one. Organized fundraising drive and donation collection for Loaves & Fishes Food Pantry.

## HONORS & AWARDS

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Pearl Hacks Hackathon Award, Qorvo Inc. (2023); Regeneron STS (Science Talent Search) Scholar (2022, top 300 of 2,162 applicants); National JSHS Momentum Grant (2022); NC International Science Challenge Winner (2022); Bowman-Brockman Endowment (2022); NASA iTech Honorable Mention & Top 10 Finalist (2021); Pete Conrad Scholar (2021); Coca-Cola Scholar Semifinalist (2021, top 2% of 68,000+ applicants); Carson Scholar (2021); AJAS (American Junior Academy of Sciences) Fellow & Lifetime Honors (2020-2021); International BioGENEius Challenge Finalist & NC Winner (2020); World Science Scholar (2019, 1 of 30 selected globally); Intel ISEF (International Science & Engineering Fair) Finalist (2019); USCF Top-34 Female Chess Player in Age Group (2019).